

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

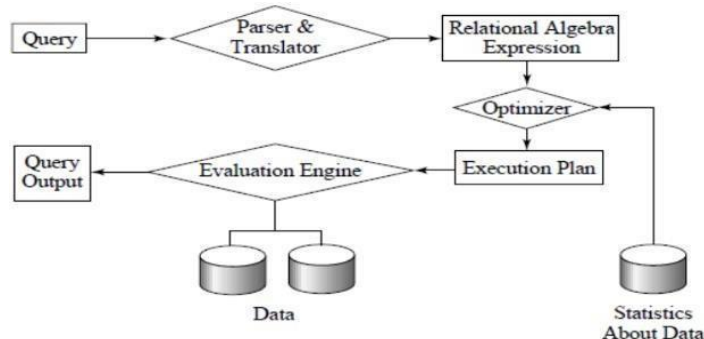
Regulation	:	2023
Course Name	:	DATABASE MANAGEMENT SYSTEMS
Course Code	:	JCS2401
Year / Semester / Branch	:	II / IV / CSE
Unit No	:	4
Unit Title	:	DATA ORGANIZATION AND QUERY OPTIMIZATION
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1.	Define the term RAID. RAID (Redundant Array of Independent Disks) is a technology that combines multiple physical disks into a single logical unit to improve performance, reliability, or both.	2	K1	CO4
2.	Outline the factors to be considered when choosing a RAID level • Monetary cost of extra disk storage requirements. • Performance requirements in terms of number of I/O operations • Performance when a disk has failed and Performances during rebuild	2	K2	CO4
3.	Define a record in a file. A record in a file is a collection of related fields that are treated as a single unit, representing a complete set of information about one specific entity. It is a fundamental data structure component, where multiple records are stored sequentially to form a file.	2	K1	CO4
4.	How does indexing work ? Indexing is a data structure technique used to speed up data retrieval by creating a separate, sorted structure that allows the	2	K2	CO4

	database to locate rows without scanning the entire table. It dramatically reduces disk I/O, enhances query performance and is crucial for handling large datasets.			
5.	List out the types of storage devices Primary storage, Secondary storage, Tertiary storage, Volatile storage, Nonvolatile storage etc.	2	K1	CO4
6.	Define an ordered index in DBMS. An ordered index in a Database Management System (DBMS) is a specialized data structure, typically a sorted list of key-value pairs (search key and pointer), that stores index entries in a sorted order to facilitate fast data retrieval. Ordered indexes enhance query performance by allowing binary search algorithms, significantly reducing the need for full table scans.	2	K1	CO4
7.	Mention the structure of B+ tree. A B+ tree is a self-balancing, multi-level tree data structure used in DBMS for indexing, optimizing search and range queries by storing all data/pointers exclusively in leaf nodes. Leaf nodes are linked sequentially for efficient traversal, while internal nodes only store keys to guide search, making them ideal for high-fanout disk-based storage.	2	K1	CO4
8.	Define a B-tree. A B-tree is a self-balancing, m-way tree data structure used in DBMS to optimize searching, insertion, and deletion of large datasets, specifically designed to minimize disk I/O operations. It maintains sorted data with all leaf nodes at the same depth, ensuring efficiency for operations while keeping the tree shallow.	2	K1	CO4
9.	Specify the function of static hashing. Static hashing is a data storage technique where a fixed number of buckets (memory locations) are used, and a consistent hash function maps search keys to specific bucket addresses. The number of buckets remains constant, making it ideal for scenarios with static, predictable data volume.	2	K1	CO4
10.	State the process of query optimization Query optimization in a Database Management System (DBMS) is the process of selecting the most efficient execution plan for a given SQL query from among all possible equivalent plans. The primary goal is to minimize the resources used (such as CPU, memory, and disk I/O) and the overall query execution	2	K1	CO4

	time to provide faster results to users.			
11.	Why is RAID important for file organization? RAID (Redundant Array of Independent Disks) is used in DBMS to improve data reliability, increase storage performance (I/O speed), and ensure high availability by distributing data across multiple disks. It protects against data loss via mirroring or parity, allowing databases to continue running despite hardware failures.	2	K2	CO4
12.	Interpret the working of sequential file organization. Sequential file organization is a fundamental method where records are stored and accessed in a linear, one-after-another sequence. This method is simple in design and particularly effective for processes that require accessing most or all records in order, such as generating payroll slips or reports.	2	K2	CO4
13.	Differentiate between Primary Index and Secondary Index: • Primary Index: □ Created on the primary key of a file. □ Data file is physically ordered based on this key. □ Only one primary index is possible. □ Usually a sparse index. • Secondary Index: □ Created on a non-key field. □ Data file is not ordered based on this field. □ Multiple secondary indexes can exist. □ Usually a dense index.	2	K2	CO4
14.	How does hashing improve search efficiency? Hashing improves search efficiency in a DBMS by providing direct, constant-time ($\Theta(1)$ on average) access to data records, eliminating the need for sequential searches or tree traversals, which are significantly slower for large databases.	2	K2	CO4
15.	Give reasons for the usage of ordered indices. • Fast Data Retrieval (Search Speed). • Efficient Range Queries	2	K2	CO4

	- Reduced Disk I/O - Optimized Sorting and Grouping - Support for Unique Constraints			
16.	How does Linear probing work? - Linear probing is an open addressing collision resolution technique where, if a hash collision occurs, the system searches for the next available slot sequentially in the hash table. - If a key hashes to an occupied position, the system checks the next slot using: $h_i(k) = (h(k) + i) \mod m$ where i = 0,1,2,3,...and m is the table size.	2	K2	CO4
17.	Interpret the limitation of static hashing The primary limitation of static hashing in DBMS is its fixed number of buckets, which cannot adapt to changing database sizes. As data grows, this causes frequent bucket overflows (collisions), leading to long overflow chains and degraded search performance, while shrinking data leads to wasted memory space.	2	K2	CO4
18.	Illustrate on dynamic hashing Dynamic hashing adjusts the number of buckets, growing or shrinking to accommodate data changes, which improves performance for dynamic datasets and minimizes overflow.	2	K2	CO4
19.	List the steps in in Query Processing and give a diagrammatic illustration . Basic steps in query processing - Parsing and translation - Optimization - Evaluation  <pre> graph TD Query[Query] --> PT{Parser & Translator} PT --> RA[Relational Algebra Expression] RA --> Opt{Optimizer} Opt --> EP[Execution Plan] EP --> EE{Evaluation Engine} EE --> QO[Query Output] Data[(Data)] --> EE Stats[(Statistics About Data)] --> EE </pre>	2	K2	CO4

20.	Why is query optimization important? Query optimization in a DBMS is crucial because it ensures that queries run efficiently by selecting the most cost-effective execution plan out of many possibilities. This process minimizes the use of system resources, resulting in faster data retrieval and overall better database performance.	2	K2	CO4
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PART B

Q.No.	Questions	Marks	BTL	CO
1	Discuss in detail how records are stored and accessed using secondary storage devices, with suitable examples.	16	K2	CO4
2	Describe the usage of RAID techniques in a storage system.	16	K2	CO4
3	Categorize the various Indexing and Hashing techniques used for file organization and explain each with a suitable example.	16	K4	CO4
4	Analyze the structure and operations of B-tree and B+ tree indexing techniques. Illustrate their differences with suitable examples.	16	K4	CO4
5	Explain in detail the different hashing techniques suitable method for efficient data retrieval.	16	K2	CO4
6	Illustrate the steps involved in query processing and describe how each step improves query performance.	16	K2	CO4

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